

B.Tech IV Sem (Electrical Engg.)
ME 24252 (Fundamentals of Design)

Note: All questions carry equal marks. Support the answers with suitable examples.

S. No.		5 Marks
1.	Select a target customer, observe their daily routine related to the problem context. Defined your problem statement by using the empathy mapping from this observation.	CO2 BT4
2.	Imagine you are the founder of Zomato, a food delivery company. Explain the main problem that Zomato wanted to solve and the solution it provided to customers. By following the steps of the design process, describe how the root cause of the problem was identified and how the actual solution was developed.	CO1 BT5
3.	A start-up recently launched a smart backpack for college students. It had features like an inbuilt charger, Bluetooth tracker, and automatic lock system. Despite being technologically impressive, sales were poor, and the company shut down in less than a year. Investors argued that the team focused too much on features and ignored student's real needs. Identify the major causes of this failure. How could the principles of Design Thinking have helped this start-up understand users better and avoid an over-engineered but under-used product ?	CO1 BT3
4.	Madhya Pradesh Metro Rail Corporation is planning to build a new Indore-Ujjain-Pithampur metro corridor. The project involves clear steps: land surveys, structural design, tunnelling, station construction, safety checks, and commissioning. Because requirements are fixed and changes are very costly once construction begins, the team suggests using the Waterfall model. Explain why the Waterfall approach is actually well-suited here. What strengths of Waterfall make it preferable over iterative methods in this type of project ?	CO3 BT4